CSA5108 Cryptography and Network security

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Exp 12 : Hill Cipher — Encryption and Decryption
Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main(void) {
   char input[500], p[500], c[500], d[500];
   int n=0;

   int K[2][2]={{9,4},{5,7}};
   int I[2][2]={{5,12},{15,25}};

   printf("Enter plaintext: ");
   fgets(input,sizeof(input),stdin);

   for(int i=0;input[i];i++)
      if(isalpha((unsigned char)input[i]))
         p[n++]=tolower((unsigned char)input[i]);

   if(n%2) p[n++]='x';
   p[n]='\0';

   for(int i=0;i<n;i+=2) {
      int x=p[i]-'a', y=p[i+1]-'a';
      c[i]=(K[0][0]*x+K[0][1]*y)%26+'a';
      c[i+1]=(K[1][0]*x+K[1][1]*y)%26+'a';
   }
   c[n]='\0';

   for(int i=0;i<n;i+=2) {
      int x=c[i]-'a', y=c[i+1]-'a';
```

```
        d[i]=(I[0][0]*x+I[0][1]*y)%26+'a';
        d[i+1]=(I[1][0]*x+I[1][1]*y)%26+'a';
    }
    d[n]='\0';

    printf("Prepared plaintext: %s\n",p);
    printf("Ciphertext: %s\n",c);
    printf("Decrypted plaintext: %s\n",d);

    return 0;
}
```

